

नेशनल इंस्टीट्यूट ऑफ इलेक्ट्रॉनिक्स एंड इंफॉर्मेशन टेक्नोलॉजी, चेन्नई

National Institute of Electronics and Information Technology, Chennai

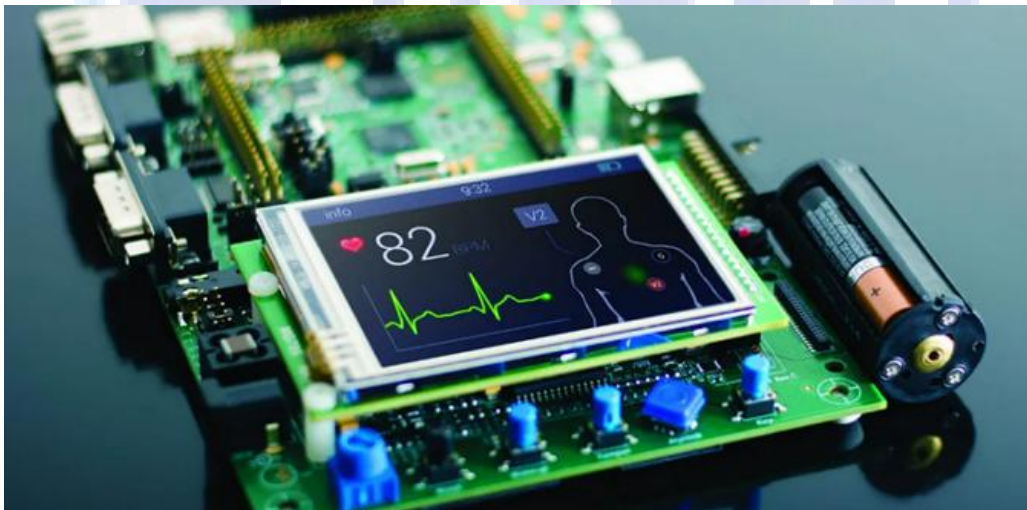
Autonomous Scientific Society of Ministry of Electronics & Information Technology (MeitY), Govt. of India

ISTE Complex, 25, Gandhi Mandapam Road, Chennai - 600025

Course Prospectus

Embedded System Junior Developer (O-Level 'Embedded System Design')

Mode: Online
(NSQF-Level 4)



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Course Prospectus

Course Name: Embedded System Junior Developer (O-Level 'Embedded System Design')

Course Code: ED151

NSQF Level: 4

Duration: 450 Hrs

Last Date of Registration: 1st Oct 2025

Course Start Date: 06th Oct 2025

Fee Details:

Registration Fee- Rs. 1000/- (Adjusted with Total Fee)

Total Fee – Rs. 24,500/- (Fee is waived for SC/ST students)

Preamble:

In today's world, embedded systems are all over, homes, offices, cars, factories, hospitals and consumer electronics. The inherent value of embedded systems lies in its pervasiveness. They are literally embedded in all electronic products, from consumer electronics to office automation, automotive, medical devices and communications. We live in the age, where information is just one click away and talking just one touch away.

The Embedded T Industry is growing rapidly with the introduction of wide variety of Product for various applications catering to different sector demands. This increases the complexity of embedded system design; currently there is a shortage of qualified engineer with good Embedded Design and Development skills. Sector will continue to grow with introduction of new innovative products & application; therefore, the need for Skilled Engineers will continue to grow. Hence, there need an advanced training program in Embedded Field, this course focuses on the architecture and programming of embedded processors, development of applications using Microcontroller/Embedded Linux Systems and porting the applications on ARM.

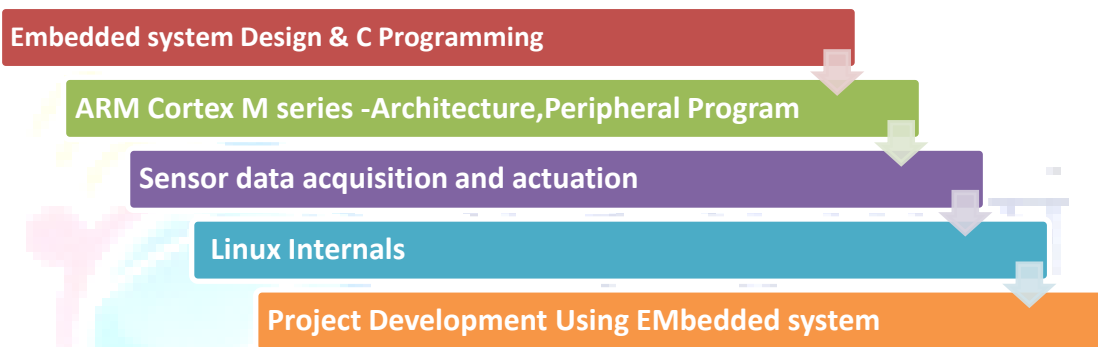
Objective of the Course:

The objective of O-Level chip design program is to introduce the participants to the necessary skills and knowledge to excel in the field of Embedded System Applications using suitable Hardware and Software tools. The program is of immediate relevance to industry and makes the participants exactly suitable for the Embedded Industry

Outcome of the Course:

After successful completion of this Course, students will be able to:

- Embedded system design involving microcontroller peripherals and hardware interfacing using C programming
- Develop Embedded Application using ARM Cortex-M series architecture and their ability to implement efficient peripheral programming
- Design and implement sensor data acquisition systems in embedded environments
- Design & Understand Linux kernel architecture and internals.
- Develop their software application with ARM, BeagleBone board, Linux kernel architecture



Full Flow of Course:

Course Structure:

This course contains totally Nine modules. After completing the first Eight modules, the students have to do a six weeks' project using any of the topics studied to earn the Embedded System Junior Developer certificate.

Code	NOS Name	Duration(in Hours)
ED151-1	Embedded System Design & C- Programming	60
ED151-2	Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming	60
ED151-3	Sensor data acquisition and actuation in embedded systems	60
ED151-4	Linux Internals	60
ED151-5	Employability Skills	60

ED151-6	Project Work/OJT	150
Total Duration		450

Course Fees

The course fee is **Rs. 24,500/-** including GST. Can be paid as a single instalment or in 2 instalments as given below,

Fee Component	SC-ST Candidate	General/OBC/EWS Candidate	Last Date
Registration Fee	Nil	Rs. 1000/- for others (Adjustable with total fee)	01 st Oct 2025
Tuition Fee (Including NSQF Registration & Examination Fee)			
1 st Instalment	Nil	11,750/-	05 th Oct 2025
2 nd Instalment	Nil	11,750/-	06 th Dec 2025
Total	Nil	24,500/-	

Registration Fee.

(Non-Refundable if candidate is selected for admission but did not join and if a candidate has applied but not eligible.)

SC/ST: Nil

Others: Rs. 1,000/- (Adjustable with Total fee for candidates)

However, the above registration fee shall be refunded in a few special cases as given below

- Candidates are eligible but not selected for admission.
- The course has been postponed, and the new date is not convenient for the student.
- Course cancelled.

Eligibility

- Completed 12th or equivalent in Science with Physics and Maths
(or)
- Completed 2nd Year of 3-Year Diploma in Electronics and Communication Engineering/
Electrical Engineering/CS/IT and allied branches after class 10th

Number of Seats: 30-Total [tentative]

Category	No. of Seats
SC (15%)	4
ST (7.5%)	2

GENERAL	24
Total	30

Note: Seats are allocated based on the merit of the Qualification.

How to Apply?

Candidates can apply online in our website <https://nva.nielit.gov.in/>. Payment towards the non-refundable registration fee can be paid through any of the following modes:

- Online transaction:
 - **Account Name:** NIELIT CHENNAI
 - **Account No:** 31185720641
 - **Bank name:** State Bank of India (SBI)
 - **Branch:** Kottur (Chennai)
 - **IFSC Code:** SBIN0001669.
- Pay through UPI Mobile Apps

Note: The Institute will not be responsible for any mistakes done by either the bank concerned or by the depositor while remitting the amount into our account

Last date of Registration: 01st Oct 2025

Registration Procedure:

All interested candidates are required to fill the Registration form online with registration fees before 01st Oct 2025 with all the necessary information.

Selection Criteria of candidates:

The selection to the course shall be based on the following criteria:

Selection of candidates will be based on their marks in the qualifying examination subject to eligibility and availability of seats.

- The Provisionally Selected Candidates will be informed regarding **selection via email**. In case of vacancy, additional selection lists will be prepared and will be intimated by email only.
- Provisionally selected candidate has to upload their document on registration portal for online verification.
 - ✓ Original Copies of Proof of Age, Qualifying Degree (Consolidated Mark sheet & Degree Certificate/Course Completion Certificate), 10th and 12th mark sheet.
 - ✓ One passport size photograph.
 - ✓ Self-attested copy of Govt. issued photo ID card.

- After document verification, selected candidates have to pay first instalment of **Rs. 11,750/-** or as applicable on or before **05th Oct 2025** by payment mode mentioned above. Selected candidates are requested to upload the proof of remittance of fee on registration portal and also send the proof of remittance of fee as email to ishant@nielit.gov.in or trng.chennai@nielit.gov.in

Admission:

All provisionally selected candidates whose documents are verified and have paid the fees (full or first instalment) and verified by the accounts section of NIELIT Chennai will get a welcome message for the provisional admission in his/her login ID provided during registration. The credentials and URL for the online portal will be shared through WhatsApp or email.

Note: Provisionally Selected Candidates have to visit NIELIT Chennai for Certificate Verification, Biometric verification & Final Examination. Otherwise, their candidature will be cancelled without any intimation.

Discontinuing the course:

- No fees (including the security deposit) under any circumstances, shall be refunded in the event of a student who have completed the process of admission or discontinuing the course in between. No certificate shall be issued for the classes attended. Only Grade Sheet will be issued.
- If candidates are not uploading consecutive 3 assignments within assigned time their candidature will be cancelled without any notice and all fees paid will be forfeited.
- If candidates are not appearing for any internal examinations/practical their candidature will be cancelled without any notice and all fees paid will be forfeited

Course Timings:

This program is a practical oriented one and hence there shall be more lab than theory classes. The cloud based online theory classes will be on forenoon and lab session will be conducted mostly on afternoon time.

Address:

National Institute of Electronics and Information Technology
ISTE Complex, No. 25, Gandhi Mandapam Road, Chennai – 600025
E-mail: ishant@nielit.gov.in
Phone: 044-24421445
Contact Person: Mr. Ishant Kumar Bajpai
Mobile: 9445240125 [Call @ 9 AM to 5 PM]

Course enquiries:

Students can enquire about the various courses either on telephone or by personal contact between 9.15 A.M. to 5.15 P.M. (Lunch time 1.00 pm to 1.30 pm) Monday to Friday.

Placement:

Students who have completed the course successfully and qualified, will be given placement guidance and career counselling to crack the interviews.

Important Dates:

- **Last Date of Registration:** 01st Oct 2025
- **Payment of first instalment fee:** on or before 05th Oct 2025
- **Course Start Date:** 06th Oct 2025
- **Payment of second instalment fee:** 08th Dec 2025

Examination & Certification:

Final Certificates will be issued after successful completion of all the modules including mini project. For getting certificate a candidate has to pass each module individually with minimum required marks of 50%.

Examination Scheme:

Examination scheme for each module is as follows:

Module Name	Total Marks	Written	Practical / Assignment
Embedded System Design & C-Programming	100	25	75
Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming	100	25	75
Sensor data acquisition and actuation in embedded systems	100	20	80
Linux Internals	100	25	75
Employability Skills	100	25	75
Project Work/OJT	100	NA	100
Total	600	120	480

Grading Scheme:

- Following Grading Scheme (on the basis of total marks) will be followed:

Grade	S	A	B	C	D	Fail
Marks Range (in %)	85 to 100	75 to 84	65 to 74	55 to 64	50 to 54	Below 50

- Final Grading as per above grading scheme will be given on the basis of total marks obtained in all modules. For last module (ES 609) grade will be given on the basis of project demonstration.

Lab Infrastructure Details:

Hardware Facilities:

- Development Boards - STM32, ARM Cortex-M4, Arduino Uno, NodeMCU, MSP430, Beagle Bone Black, Raspberry Pi
- Sensors–PIR, Ultrasonic, Soil Moisture, Accelerometer & Gyro meter
- Camera Module, Sense Hat, Capacitive Touch Screen
- Wireless Sensor Network Radio and Related Modules with Integrated Antenna
- Aardvark I2C/SPI Host adapter, I2C/SPI development board, CAN Development board, Komodu CAN Duo Interface

Software Facilities:

- C Programming IDE's Dev C++
- Keil Uvision, STM32 CubeMx, Stm32Cube IDE
- Ubuntu Linux system
- Proteus VSM

Faculty Profile:



Ishant Kumar Bajpai (Scientist D), extensive experience spans over a decade and revolves around implementing funded projects in two key areas: the Internet of Things (IoT) and Very Large Scale Integration (VLSI) with applications in the Biomedical and Automotive industries. His research interests are focused on FPGA implementations of Digital Signal Processing (DSP) systems.



RAGHAVENDRA S – Project Engineer

expertise in Embedded System Design and IoT Training, Robotics and automation and handling training session and Labs on UiPath studio with the background of electronics and instrumentation. Skilled in embedded system development.



PRASANTH B R Resource Person – IoT

having 6 years of experience with skill sets of C, Embedded C, ARM Cortex Microcontrollers, Embedded Linux, RTOS etc with the background of electronics and communication.

Detailed Syllabus of the Course

ED151-1: Embedded System Design & C- Programming

Duration: 60 Hours

Objective:

To set the required background in embedded system concepts, Embedded 'C' language such as Memory management, Pointers, Data structures and architecture of the ARM Cortex processor for highly deterministic real-time applications.

Module Description:

C' and Embedded-C

- Introduction to 'C' programming
- Storage Classes
- Data Types
- Controlling program flow
- Arrays
- Functions
- Memory Management
- Pointers
- Arrays and Pointers

- Pointer to Functions and advanced topics on Pointers
- Structures and Unions
- Data Structures
- Linked List
- Stacks, Queues
- Conditional Compilation
- Pre-processor directives
- File operations
- Bitwise operations
- Typecasting

Embedded Concepts

- Introduction to embedded systems
- Application Areas
- Categories of embedded systems
- Overview of embedded system architecture
- Specialties of embedded systems
- Recent trends in embedded systems
- Architecture of embedded systems
- Hardware architecture
- Software architecture
- Application Software
- Communication Software
- Development and debugging Tools

ED151-2: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming

Duration: 60 Hours

Objective:

The objective is used to provide good understanding of Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming

Module Description:

Introduction to ARM Cortex

- Architecture Introduction to 32-bit Processors
- The ARM Architecture
- Overview of ARM
- Overview of Cortex Architecture

- Cortex M4 Register Set and Modes
- Cortex M4 Processor Core
- Data Path and Instruction Decoding
- ARM Cortex M4 Development Environment
- Assembler and Compiler
- Linkers and Debuggers
- ARM-Thumb & Thumb2 instructions
- Mixing ARM & Thumb Instructions
- Memory hierarchy
- Memory Mapping
- Cache

Cortex M4 Microcontrollers & Peripherals

- Cortex M4 based controller architecture
- Memory mapping, Cortex M4 Peripherals – RCC
- GPIO
- Timer, System timer
- UARTs, LCD, ADC & PWM
- Cortex M4 interrupt handling – NVIC
- Application development with Cortex M4 controllers using standard peripheral libraries

ED151-3: Sensor data acquisition and actuation in embedded systems

Duration: 60 Hours

Objective:

The objective of the course is to provide a thorough understanding about and hands-on practice Sensor data acquisition and actuation in embedded systems

Description:

ED151- 4: Linux Internals

Duration: 60 Hours

Objective:

To Skilling the students in Configure, Deploying and Debugging the Linux OS onto a Target Board to build a complete Embedded Product using Linux Kernel.

Description:

Architecture of Embedded Linux:

- Basic Operating System Concepts
- History& Benefits of Linux
- Fundamentals of Embedded Linux OS
- What is Kernel?
- Task of kernels
- Types of kernels
- Kernel Architecture Overview
 - User Space
 - Kernel Space
- Kernel Functional Overview
 - File System
 - Process Management
 - Address Spaces and Privilege Levels
 - Memory Management
 - System Calls

Configuring the Linux Environment:

- Linux environment
- Types of Hosts
- Types of Host/Target Development Setups
- Types of Host/Target Debug Setups
- Embedded Environment Tools
- GNU Tool-chain Cross Compilers

Tool-chain: Configuration and Cross-Compilation:

- What is a tool-chain?
- Native vs. cross-compilation
- Toolchain Components
- Toolchain choices
- Using build root to build the toolchain
- Configuration options
- Adding path variables to startup scripts (.bashrc)
- The CROSS_COMPILE variable
- Validating the cross-compiler

Linux Bootloader & U-Boot:

- Boot-loader Phases
- U-boot – Embedded boot loader
- What does u-boot do?
- Navigating the u-boot sources
- Configuring and Cross-compiling u-boot
- Installing u-boot on the target
- Understanding u-boot commands
- Changing environment variables to setup kernel booting

- Transferring files to the target using tftp

Embedded Linux Kernel:

- Kernel Features
- Kernel Subsystems ▪
- Memory Manager
- Scheduler
- Embedded Storage
- I/O Subsystem
- Network Subsystem
- Navigating the kernel sources
- Kernel Configuration
- Kernel Compilation
- Booting the kernel using u-boot
- Module compilation and Installation to RootFS
- Procedure for adding a new driver to the kernel
- Applying patches

Building Root File System:

- Introduction to File system
- Linux directory structure
- Organization and Important directories /dev file system
- What next after kernel booting
- init and startup scripts
- Downloading & Compiling RootFS
- RootFS in Flash/SD Card Storage

Porting OS in ARM Board:

- Kernel Compilation
- Booting the kernel using u-boot
- Porting Linux in ARM Board

ED151-5: Employability Skill

Duration: 60 Hours

Description:

Constitutional Values – Citizenship:

- Explain constitutional values, including civic rights and duties, citizenship, responsibility towards society, and personal values and ethics such as honesty, integrity, caring, and respecting others that are required to become a responsible citizen.

- Demonstrate how to practice different environmentally sustainable practices.
- Becoming a Professional in the 21st Century.
- Discuss relevant 21st-century skills required for employment.
- Highlight the importance of practicing 21st-century skills like Self-Awareness, Behavior Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn, etc., in personal or professional life.
- Create a pathway for adopting a continuous learning mindset for personal and professional development.

Basic English Skills:

- Show how to use basic English sentences for everyday conversation in different contexts, in person and over the telephone.
- Read and understand text written in basic English.
- Write a short note/paragraph/letter/e-mail using correct basic English.
- Career Development & Goal Setting
- Create a career development plan.
- Identify well-defined short- and long-term goals.

Communication Skills:

- Demonstrate how to communicate effectively using verbal and nonverbal communication etiquette.
- Write a brief note/paragraph on a familiar topic.
- Explain the importance of communication etiquette, including active listening for effective communication.
- Role-play a situation on how to work collaboratively with others in a team.
- Diversity and Inclusion
- Demonstrate how to behave, communicate, and conduct appropriately with all genders and PwD.
- Discuss the significance of escalating sexual harassment issues as per the POSH act.

Financial and Legal Literacy:

- Discuss various financial institutions, products, and services.
- Demonstrate how to conduct offline and online financial transactions safely and securely and check passbook/statement.
- Explain the common components of salary such as Basic, PF, Allowances (HRA, TA, DA, etc.), tax deductions.
- Calculate income and expenditure for budgeting.
- Discuss legal rights, laws, and aids.

Essential Digital Skills:

- Describe the role of digital technology in day-to-day life and the workplace.
- Demonstrate how to operate digital devices and use the associated applications and features safely and securely.

- Demonstrate how to connect devices securely to the internet using different means.
- Follow the dos and don'ts of cybersecurity to protect against cybercrimes.
- Discuss the significance of displaying responsible online behavior while using various social media platforms.
- Create an email id and follow email etiquette to exchange emails.
- Show how to create documents, spreadsheets, and presentations using appropriate applications.
- Utilize virtual collaboration tools to work effectively.

Entrepreneurship:

- Explain the types of entrepreneurship and enterprises.
- Discuss how to identify opportunities for potential business, sources of funding, and associated financial and legal risks with its mitigation plan.
- Describe the 4Ps of Marketing-Product, Price, Place, and Promotion and apply them as per requirement.
- Create a sample business plan for the selected business opportunity.

Customer Service:

- Classify different types of customers.
- Demonstrate how to identify customer needs and respond to them in a professional manner.
- Discuss various tools used to collect customer feedback.
- Discuss the significance of maintaining hygiene and dressing appropriately.

Getting ready for Apprenticeship & Jobs:

- Draft a professional Curriculum Vitae (CV).
- Use various offline and online job search sources to find and apply for jobs.
- Discuss the significance of maintaining hygiene and dressing appropriately for an interview.
- Role-play a mock interview.
- List the steps for searching and registering for apprenticeship opportunities.

ED151-6: Project Work/OJT

Duration: 150 Hours

Description:

The students can select Embedded system projects. The project can be implemented using ARM, RPi, BBB etc which students have studied and used during the course.
